

MAT 535 HW08

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Problem 1 (§13.3#4). The construction of the regular heptagon amounts to the constructibility of $\alpha = \cos(2\pi/7)$. We shall see later that α satisfies the equation $x^3 + x^2 - 2x + 1 = 0$. Use this to prove that the regular heptagon is not constructible by straightedge and compass.

Solution. The polynomial $x^3 + x^2 - 2x + 1$ is irreducible because it has no rational roots (the only candidate rational roots are ± 1 by the rational root theorem, and they aren't roots). This means $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ is not a power of 2, hence α is not constructible by straightedge and compass by Proposition 23. $\square$

Problem 2 (§13.3#5). Use the fact that $\alpha = 2\cos(2\pi/5)$ satisfies the equation $x^2 + x - 1 = 0$ to conclude that the regular pentagon is constructible by straightedge and compass.

Solution. Since $\mathbb{Q}(\alpha)$ is a quadratic extension of $\mathbb{Q}$, α is constructible, and then $\sin(2\pi/5)$ can also be constructed (since $[\mathbb{Q}(\sqrt{1-\alpha^2}) : \mathbb{Q}(\alpha)] \leq 2$). Then, we can rotate segments by $2\pi/5$, so we can also rotate segments by $\pi - 2\pi/5 = 3\pi/5$, which is the measure of the interior angles of a regular pentagon. Draw a segment A_0A_1 and, for $i \geq 0$, let A_{i+2} be the rotation A_i about A_{i+1} by $3\pi/5$ counterclockwise; this constructs the regular pentagon $A_0A_1A_2A_3A_4$. $\square$

Problem 3 (§14.5#2). Determine the subfields of $\mathbb{Q}(\zeta_8)$ generated by the periods of ζ_8 and in particular show that not every subfield has such a period as a primitive element.

Solution. The Galois group $G = \text{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ has three nontrivial subgroups $\{1, \sigma_3\}$, $\{1, \sigma_5\}$, and $\{1, \sigma_7\}$, where $\sigma_a \in G$ is the automorphism sending ζ_8 to ζ_8^a . Thus the periods of ζ_8 are

$$\zeta_8 + \zeta_8^3 = i\sqrt{2}, \quad \zeta_8 + \zeta_8^5 = 0, \quad \zeta_8 + \zeta_8^7 = \sqrt{2}.$$

The corresponding subfields are respectively $\mathbb{Q}(i\sqrt{2})$, $\mathbb{Q}$, and $\mathbb{Q}(\sqrt{2})$, so the subfield $\mathbb{Q}(i)$ does not have a period of ζ_8 as a primitive element. $\square$

Problem 4 (§14.5#5). Let p be a prime and let S_n be the sum of the n th powers of all the primitive p th roots of unity. Show that $S_n = -1$ when p does not divide n and $S_n = p-1$ when p divides n .

Solution. Let ζ be a primitive p th root of unity, so that $S_n = \zeta^n + \zeta^{2n} + \dots + \zeta^{(p-1)n}$. If p divides n then $\zeta^n = 1$, so of course $S_n = 1$. If p does not divide n , then multiplication by $\bar{n}$ (the residue class of n) is an isomorphism on $(\mathbb{Z}/p\mathbb{Z})^\times$, so

$$S_n = \zeta^n + \zeta^{2n} + \dots + \zeta^{(p-1)n} = \zeta + \zeta^2 + \dots + \zeta^{p-1} = S_1$$

because the exponents on the left are a permutation of the exponents on the right (modulo p). We can then see that $S_1 = -1$ by Vieta's formulas applied to $\Phi_p(x) = x^{p-1} + x^{p-2} + \dots + 1$. $\square$

Problem 5 (§14.5#7). Show that complex conjugation restricts to the automorphism $\sigma_{-1} \in \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ of the cyclotomic field of n th roots of unity. Show that the field $K^+ = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$ is the subfield of real elements of $K = \mathbb{Q}(\zeta_n)$, called the **maximal real subfield** of K .

Solution. The conjugate of ζ_n is ζ_n^{-1} , and the only automorphism of $\text{Gal}(K/\mathbb{Q})$ sending ζ_n to ζ_n^{-1} is σ_{-1} , so the conjugation automorphism must restrict to σ_{-1} . The maximal real subfield L the fixed field of the order-2 subgroup $\{1, \sigma_{-1}\}$, hence $[K : L] = 2$. Since $\zeta_n + \zeta_n^{-1}$ is real, $K^+ \subseteq L$, hence $[K : K^+] \geq 2$. Letting $\eta = \zeta_n + \zeta_n^{-1}$ and noting that $\zeta_n \zeta_n^{-1} = 1$, we can see that ζ_n is a root of the quadratic polynomial,

$$p(t) = t^2 - \eta t + 1$$

so $[\mathbb{Q}(\zeta_n) : K^+] = [K : K^+] \leq 2$, hence $[K : K^+] = [K : L] = 2$ and $K^+ \subseteq L$, implying $K^+ = L$. $\square$